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Análisis e interpretación de datos de alto rendimiento

1. Command Line (bash - shell introduction)

On the next two lectures we are going to use command line. You can start visualizing the following tutorials:

Cheatsheet:

https://devhints.io/bash

Tutorial:

https://www.learnshell.org/

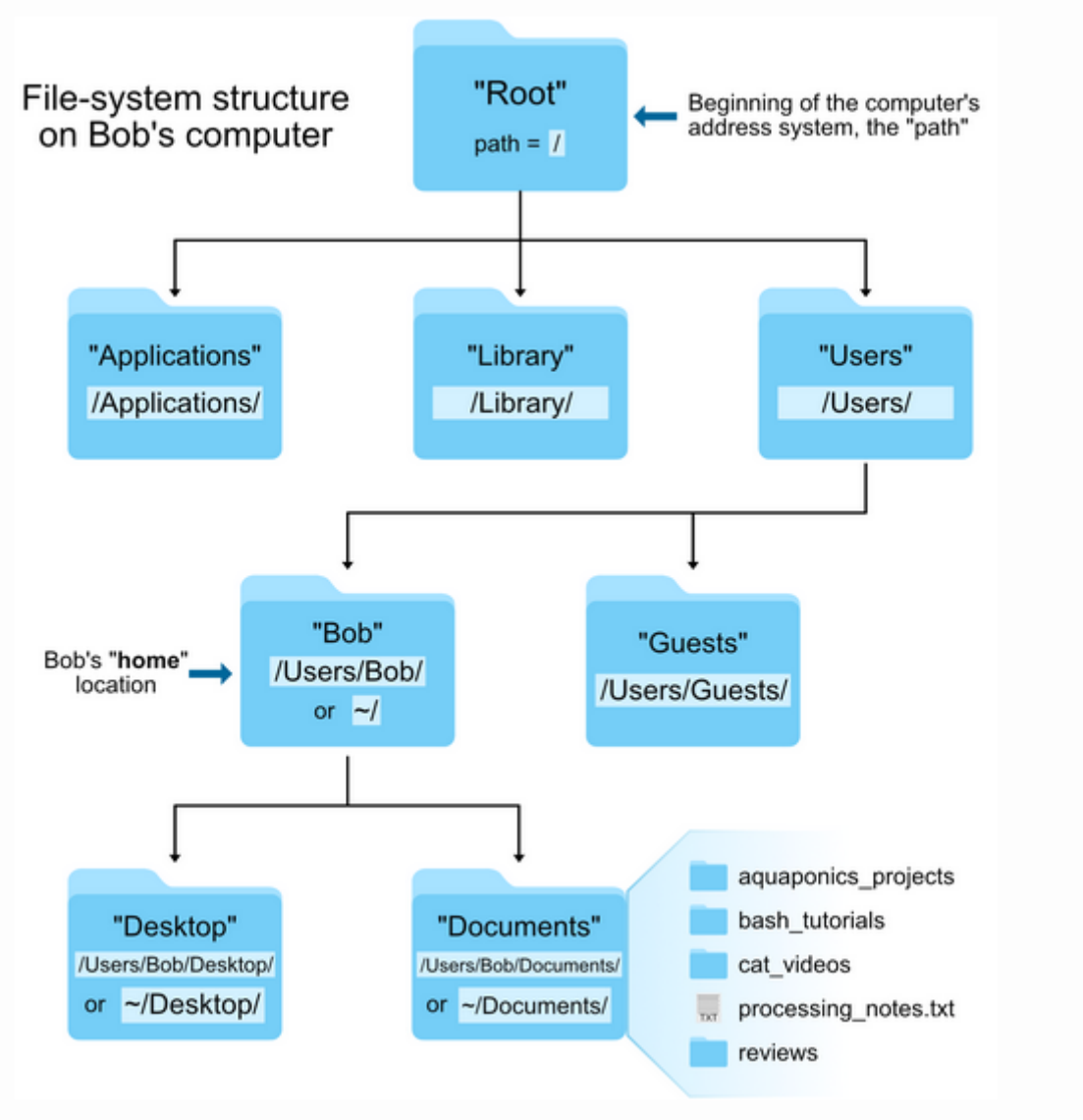
** Complete book:**

https://www.tldp.org/LDP/Bash-Beginners-Guide/Bash-Beginners-Guide.pdf

More advanced tutorial:

https://guide.bash.academy/

Linux system structure



Some important commands

Command	Function
`date`	prints the time and date
`pwd`	prints where you are in the computer (print working directory)
`ls`	lists contents of a directory (list)
`cd`	change directories
`head`	prints the first few lines of a file
`tail`	prints the last few lines of a file
`less`	allows you to browse a file (exit with "q" key)
`wc`	count lines, words, and characters in a file
`cp`	copy a file or directory (use with caution)
`mv`	mv a file or directory (use with caution)
`rm`	delete a file or directory (use with caution)
`mkdir`	create a directory
`rmdir`	delete an empty directory
`nano`	create and edit plain text files at the command line

shell pipes

```
pwd          # Print Working Directory
ls -l        # List files
cd           # Change Directory
cp           # Copy file
mv           # Move file

# Change directory
cd Desktop/course_materials/

# Display tree structure
tree

# List current folder
ls -l
ll         # short version of ls -l
```

```
# list recursively (-r) all the files in course_materials subfolders
ls -l -r course_materials/*
```

You should see something similar to this:

```
bash-4.2$ ls -l -r *ls -l -r course_materials/*
course_materials/genomes:
total 4
drwxr-xr-x 2 ri001 ubcg71 4096 Mar 12 18:44 wholeGenomeExamples
drwxr-xr-x 3 ri001 ubcg71  217 Mar 12 18:44 annotation
drwxr-xr-x 2 ri001 ubcg71   78 Mar 12 18:44 AFPN02.1

course_materials/fastq:
total 612148
-rw-r--r-- 1 ri001 ubcg71 626838750 Mar 12 18:44 final_merge_syntetic_reads.fq.gz

course_materials:
total 0
drwxr-xr-x 5 ri001 ubcg71 146 Mar 12 18:44 genomes
drwxr-xr-x 2 ri001 ubcg71  54 Mar 12 18:44 fastq
```

First steps on bash

```
# Print on screen. Writes its arguments to the standard output
echo "My first echo"

# Change directory
cd /workspace/NGS_practicals

# Command line editor (https://www.cs.colostate.edu/helpdocs/vi.html)
#  i to edit
#  ESC and then ":q" to quit
#  ESC and then ":x" to save and exit
vi test.txt

# Read .gz compressed file        zless
zless merge_syntetic_reads.fq.gz
```

Bash text editors

Vi

First steps with **Vi** editor.

Vi manual https://www.cs.colostate.edu/helpdocs/vi.html

```
# Change directory to student folder and pwd
cd /workspace/NGS_practicals
pwd

# Create test folder
mkdir test

# Change directory to just created folder and pwd
cd test
pwd

# Create test_file.txt and edit in bash with vi.
vi my_first_file.txt
```

Short vi editor cheatsheet:

Press	to
i	edit
ESC	exit insert mode
ESC and then :q"	quit
ESC and then :x	save and quit

Vi Reference Card

Modes

Vi has two modes: insertion mode, and command mode. The editor begins in command mode, where cursor movement and text deletion and pasting occur. Insertion mode begins upon entering an insertion or change command. [ESC] returns the editor to command mode (where you can quit, for example by typing `:q!`). Most commands execute as soon as you type them except for "colon" commands which execute when you press the return key.

Quitting

exit, saving changes	`:x`
quit (unless changes)	`:q`
quit (force, even if unsaved)	`:q!`

Inserting text

insert before cursor, before line	`i , I`
append after cursor, after line	`a , A`
open new line after, line before	`o , O`
replace one char, many chars	`r , R`

Motion

left, down, up, right	`h , j , k , l`
next word, blank delimited word	`w , W`
beginning of word, of blank delimited word	`b , B`
end of word, of blank delimited word	`e , E`
sentence back, forward	`( , )`
paragraph back, forward	`{ , }`
beginning, end of line	`0 , $`
beginning, end of file	`1G , G`
line *n*	*n*`G` or `:`*n*
forward, back to char *c*	`fc , Fc`
forward, back to before char *c*	`tc , Tc`
top, middle, bottom of screen	`H , M , L`

Deleting text

Almost all deletion commands are performed by typing `d` followed by a *motion*. For example `dw` deletes a word. A few other deletions are:

character to right, left	`x , X`
to end of line	`D`
line	`dd`
line	`:d`

Yanking text

Like deletion, almost all yank commands are performed by typing `y` followed by a *motion*. For example `y$` yanks to the end of line. Two other yank commands are:

line	`yy`
line	`:y`

Changing text

The change command is a deletion command that leaves the editor in insert mode. It is performed by typing `c` followed by a *motion*. For example `cw` changes a word. A few other change commands are:

to end of line	`C`
line	`cc`

Putting text

put after position or after line	`p`
put before position or before line	`P`

Registers

Named registers may be specified before any deletion, change, yank, or put command. The general prefix has the form `"c` where `c` may be any lower case letter. For example, `"adw` deletes a word into register `a`. It may thereafter be put back into the text with an appropriate put command, for example `"ap`.

Markers

Named markers may be set on any line of a file. Any lower case letter may be a marker name. Markers may also be used as the limits for ranges.

set marker *c* on this line	`mc`
goto marker *c*	`'c`
goto marker *c* first non-blank	`'c`

Search for strings

search forward	`/`*string*
search backward	`?`*string*
repeat search in same, reverse direction	`n , N`

Replace

The search and replace function is accomplished with the `:s` command. It is commonly used in combination with ranges or the `:g` command (below).

replace pattern with string	`:s/`*pattern*`/`*string*`/`*flags*
flags: all on each line, confirm each	`g , c`
repeat last :s command	`&`

Regular expressions

any single character except newline	`.` (dot)
zero or more repeats	`*`
any character in set	`[...]`
any character not in set	`[^ ...]`
beginning, end of line	`^ , $`
beginning, end of word	`\< , \>`
grouping	`\(...\)`
contents of *n*th grouping	`\`*n*

Counts

Nearly every command may be preceded by a number that specifies how many times it is to be performed. For example `5dw` will delete 5 words and `3fe` will move the cursor forward to the 3rd occurance of the letter `e`. Even insertions may be repeated conveniently with this method, say to insert the same line 100 times.

Ranges

Ranges may precede most "colon" commands and cause them to be executed on a line or lines. For example `:3,7d` would delete lines 3–7. Ranges are commonly combined with the `:s` command to perform a replacement on several lines, as with `:.,$s/pattern/string/g` to make a replacement from the current line to the end of the file.

lines *n*–*m*	`:`*n*`,`*m*
current line	`:.`
last line	`:$`
marker *c*	`:'c`
all lines	`:%`
all matching lines	`:g/`*pattern*`/`

Files

write file (current file if no name given)	`:w` *file*
append file (current file if no name given)	`:w >>`*file*
read file after line	`:r` *file*
read program output	`:r !`*program*
next file	`:n`
previous file	`:prev`
edit new file	`:e` *file*
replace line with program output	`:.!`*program*

Other

toggle upper/lower case	`~`
join lines	`J`
repeat last text-changing command	`.`
undo last change, all changes on line	`u , U`

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Nano

First steps with **Nano** editor.

```
# Text editor in bash. NOTE Do not alter them !!!
nano test_file.txt

# ^        means "ctrl"
# ^X       exit
```

Nano Cheat Sheet Version 2.7

```
^G  (F1)        Display this help text
^X  (F2)        Close the current file buffer / Exit from nano
^O  (F3)        Write the current file to disk
^R  (F5)        Insert another file into the current one

^W  (F6)        Search for a string or a regular expression
^\  (M-R)       Replace a string or a regular expression
^K  (F9)        Cut the current line and store it in the cutbuffer
^U  (F10)       Uncut from the cutbuffer into the current line

^J  (F4)        Justify the current paragraph
^T  (F12)       Invoke the spell checker, if available
                Invoke the linter, if available
                Invoke formatter, if available

^C  (F11)       Display the position of the cursor
^_  (M-G)       Go to line and column number

^Y  (F7)        Go one screenful up
^V  (F8)        Go one screenful down
M-\ (M-|)       Go to the first line of the file
M-/ (M-?)       Go to the last line of the file

M-W (F16)       Repeat the last search
M-]             Go to the matching bracket
^^  (M-A)       Mark text starting from the cursor position
M-^ (M-6)       Copy the current line and store it in the cutbuffer

M-}             Indent the current line
M-{             Unindent the current line

M-U             Undo the last operation
M-E             Redo the last undone operation

^B  (Left)      Go back one character
^F  (Right)     Go forward one character
M-Space         Go back one word
^Space          Go forward one word
^A  (Home)      Go to beginning of current line
^E  (End)       Go to end of current line
^P  (Up)        Go to previous line
^N  (Down)      Go to next line

M-7             Go to previous block of text
M-8             Go to next block of text
M-( (M-9)       Go to beginning of paragraph; then of previous paragraph
M-) (M-0)       Go just beyond end of paragraph; then of next paragraph
M-- (M-_)       Scroll up one line without scrolling the cursor
M-+ (M-=)       Scroll down one line without scrolling the cursor

M-< (M-,)       Switch to the previous file buffer
M-> (M-.)       Switch to the next file buffer
```

```
M-V             Insert the next keystroke verbatim
^I  (Tab)       Insert a tab at the cursor position
^M  (Enter)     Insert a newline at the cursor position

^D    (Del)     Delete the character under the cursor
^H    (Bsp)     Delete the character to the left of the cursor
                Cut backward from cursor to word start
                Cut forward from cursor to next word start
                Cut backward from cursor to word start
                Cut forward from cursor to next word start
M-T             Cut from the cursor position to the end of the file

M-J             Justify the entire file
M-D             Count the number of words, lines, and characters
^L              Refresh (redraw) the current screen
^Z              Suspend the editor (if suspension is enabled)

M-3             Comment/uncomment the current line or marked lines

                Save file without prompting

                Search next occurrence backward
                Search next occurrence forward

M-X             Help mode enable/disable
M-C             Constant cursor position display enable/disable
M-O             Use of one more line for editing enable/disable
M-S             Smooth scrolling enable/disable
M-$             Soft wrapping of overlong lines enable/disable
M-P             Whitespace display enable/disable
M-Y             Color syntax highlighting enable/disable

M-H             Smart home key enable/disable
M-I             Auto indent enable/disable
M-K             Cut to end enable/disable
M-L             Hard wrapping of overlong lines enable/disable
M-Q             Conversion of typed tabs to spaces enable/disable

M-B             Backup files enable/disable
M-F             Reading file into separate buffer enable/disable
M-M             Mouse support enable/disable
M-N             No conversion from DOS/Mac format enable/disable
M-Z             Suspension enable/disable
```

Remove test folder

```
# Remove test file
rm my_first_file.txt

# Move up one directory
cd ..
# Move up two directories
cd ../..

# Remove directory. NOTE -r will do recursively
rm -r test/
```

Set up your directory structure and working environment

We will set a shortcut for the path where we keep the practical work for this set of sessions. We will then create a new directory for this practical and work in that directory. If you have copied the course_materials as suggested in the last practical, you should be able to start the practical. There is one ore thing you can do to

make life easier: set a shell variable to point to the parent directory of the course_materials so you don't have to type the full path. The following will work on the default shell, elsewhere and and on GitPod:

```
# Set the path to your workspace
st_path="/workspace/NGS_practicals"

# The order of conda channels is important! Please make sure that you have
configured your conda channels prior to installing anything with BioConda:
conda config --add channels defaults
conda config --add channels bioconda
conda config --add channels conda-forge
conda config --set channel_priority strict

# Create MultiQC conda environment for MultiQC
mamba env create -f "${st_path}"/env/.environment_multiqc.yaml
conda deactivate

# Create and load conda environment with all the tools needed to carry on
this practical.
mamba env create -f "${st_path}"/env/.environment_NGS.yaml
conda activate env_NGS
```

An introduction to NGS practical 1

This document was written to help students understand the context of the second NGS practical.

2. Background and aim of this practical

We supplied file of NGS genomic reads (*final_merge_synthetic_reads.fq.gz*). This file is in FASTQ format and was created for this practical. The reads are not totally artificial, in the sense that they originate from a real *E. coli* genome but I manipulated the reads for didactic purposes. The practical proceeds to examine and manipulate this file with the aim of aligning those reads to a reference genome (also provided).

3. Quality control of raw FASTQ files

The first step in NGS analysis is always quality control of the raw data available to you. In this case, FastQC is used to produce a report of the quality of the reads in the initial FASTQ file.

Remember that the FASTQ format is the most common format for raw NGS data. It is described in detail here: https://en.wikipedia.org/wiki/FASTQ_format . Remember that fastq files are organised in groups of 4 lines, like the extract shown below:

Fastq format

fastq format is the most common format to get the sequencing data back.

More info on:

https://en.wikipedia.org/wiki/FASTQ_format

```
manager@bl8vbox[~/Desktop/course_materials/fastq]
@AFPN02.1_merge-1076320
AGTAGTNNNNNAAGGTCGCTAATCTCTTTACGCAGATTTTTTATTCCTTCAACTAACAGCGNNNN
+
HKIHJH#####BACCCGFGGGGGGGGGG1GGEGGG0GBGG1GGGGGGGG<GGGGGGGGBCG####
@AFPN02.1_merge-1076319
AGTAGTNNNNNCTTTATCCCGAAAGCGTTTGGTAGCTCGCTGGCATTAACGGGTTCGCCAGNNNN
+
KIIKGK#####BBCCB=G>GGGGGE>GGGGGGG@GGGGDGGGGGGGFGGGGGGGGG@GGGG####
@AFPN02.1_merge-1076318
AGTAGTNNNNNTCTTGCCGATTTCCGCGTTCGGCGCGAGGCGGGTGATCATCTCCAGCACCNNNN
+
KGJKJG#####BCCCCDGGGGGGGGGGGFGGGGGGCGG0>GGG1EGGCGG@GBGDGG/G0G####
@AFPN02.1_merge-1076317
AGTAGTNNNNNTCACCGAGAAGAAGATATGTAATTCGCCAGAGAGCGACGGCCCTTGCTGANNNN
+
KJKHHH#####BCBCCGGG/GG1GGGGGGGFGGGGGGG<GGDEG1GGGGGGEG>GGGGGGG####
@AFPN02.1_merge-1076316
AGTAGTNNNNNGTTTACTATCTCGCTGCTGTTGTCGCTGGTATCGATTGTGGTGATCATGANNNN
+
GKIHHH#####C3C=C;GGGGGFGCF;DGGGGGGGGGGG@GGGGFGGGGGGDGGGGGGG1G####
```

fastq header starts with @
sequence
separator +
quality

fastq Phred quality expresed in ASCII

where the quality scores correspond to the following ASCII:

```
 SSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSSS.....................................................
 ..........................XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX......................
 ...............................IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII......................
 ..................................JJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ......................
 LLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLLL.....................................................
 !"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~
 |                         |    |        |                              |                     |
33                        59   64       73                            104                   126
 0........................26...31.......40
                          -5....0........9.............................40
                                0........9.............................40
                                   3.....9...............................41
 0.2......................26...31........41

S - Sanger        Phred+33,  raw reads typically (0, 40)
X - Solexa        Solexa+64, raw reads typically (-5, 40)
I - Illumina 1.3+ Phred+64,  raw reads typically (0, 40)
J - Illumina 1.5+ Phred+64,  raw reads typically (3, 41)
    with 0=unused, 1=unused, 2=Read Segment Quality Control Indicator (bold)
    (Note: See discussion above).
L - Illumina 1.8+ Phred+33,  raw reads typically (0, 41)
```

Download and explore our fastq file

We had improved the initial fastq file. Please, use this version on all the practical:

https://www.dropbox.com/s/91iqhvmd3ckysju/final_merge_syntetic_reads.fq.gz

Check the first few lines of your own fastq file:

```
# Change directory to fastq folder
mkdir -p "${st_path}"/course_materials/fastq/
cd "${st_path}"/course_materials/fastq/


# Download fastq file
wget
"https://www.dropbox.com/s/91iqhvmd3ckysju/final_merge_syntetic_reads.fq.gz
" -O final_merge_synthetic_reads.fq.gz

# Examine .gz compress fastq file with `zless`. You can advance pages by
pressing "space bar". NOTE exit pressing "q" quit!
zless final_merge_synthetic_reads.fq.gz

# Extract reads. Realise that after "gunzip" command the file doesn't end
in ".gz".
gunzip final_merge_synthetic_reads.fq.gz

# Examine fastq file with less. Since the file is not compressed anymore we
can see it with simple "less"" NOT "zless". NOTE exit pressing "q" quit
less final_merge_synthetic_reads.fq
```

Quality control of initial file

The first step in NGS analysis is always perform a quality control of the raw data. In this case, **FastQC** is used to produce a report of the the initial fastq file.

FastQC FAQ:

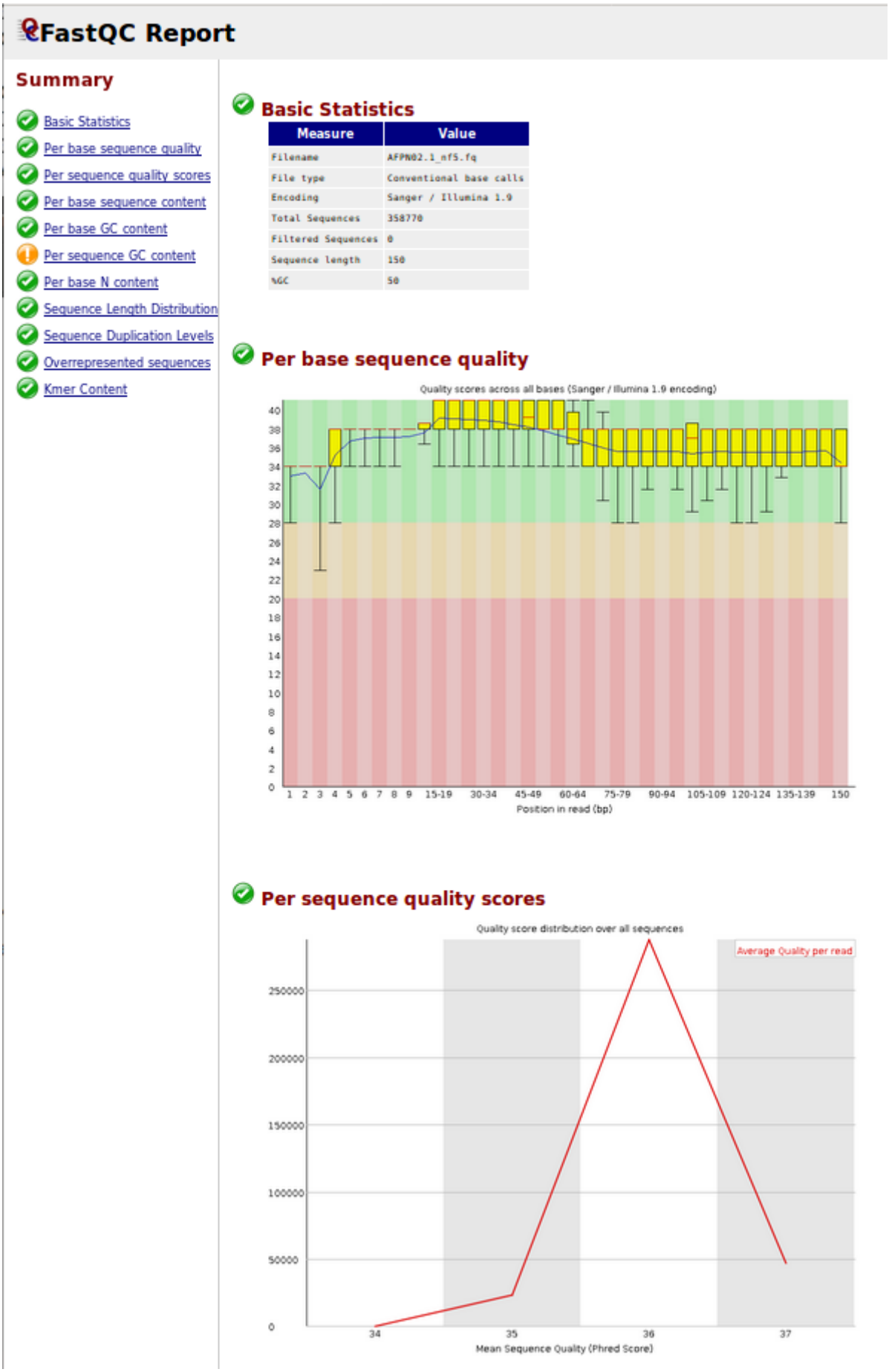
FastQC video explaining all the captured statistics https://www.youtube.com/watch?v=bz93ReOv87Y

FastQC video

Good Illumina data
https://www.bioinformatics.babraham.ac.uk/projects/fastqc/good_sequence_short_fastqc.html

Bad Illumina data https://www.bioinformatics.babraham.ac.uk/projects/fastqc/bad_sequence_fastqc.html

FastQC report



Creating our own fastQC report

```
# Change directory
cd "${st_path}"/course_materials/fastq/

#####################################################
# FastQC sequencing quality check
#####################################################

# Run FastQC
fastqc final_merge_synthetic_reads.fq

# Visualise fastQC report in firefox web-browser. NOTE exit with "ctr + c"
from the command line.
firefox final_merge_synthetic_reads_fastqc.html
```

4. Sample de-multiplexing using DNA barcodes

The FastQC report suggests that there are artificial sequences at the 5' end of these reads, something that can easily verified by checking the reads in the original fastq file. You can most likely work out what these artificial sequences are but sometimes they are given to you; We are providing you here with the barcodes used to separate the samples that have been sequenced in the same lane (this is usually done to save money). There are essentially 4 samples that have been mixed together in the raw data that was provided to you. We give you the barcodes in the guidelines of the practical and calls them **"Long"**, **"Positive"**, **"Negative"**, and **"BQ"**.

In this part of the practical, you are supposed to separate the four samples into four fastq files using the program `cutadapt` and at the same time you are using this program to trim these barcodes off the reads so as to allow mapping of the reads to the reference genome (later on).

Demultiplex barcode sequencing data with Cutadapt.

Cutadapt removes adapter sequences from high-throughput sequencing reads, demultiplex, process, ** quality trimming** and modify reads.

NOTE: **We can remove bad quality nucleotides with Cutadapt!!**

Below is a summary of the main options in cutadapt:

```
Cutadapt main parameters

Finding adapters::
  Parameters -a, -g, -b specify adapters to be removed from each read
  (or from the first read in a pair if data is paired). If specified
  multiple times, only the best matching adapter is trimmed (but see the
  --times option). When the special notation 'file:FILE' is used,
  adapter sequences are read from the given FASTA file.

  -a ADAPTER, --adapter=ADAPTER
                      Sequence of an adapter ligated to the 3' end (paired
```

```
                        data: of the first read). The adapter and subsequent
                        bases are trimmed. If a '$' character is appended
                        ('anchoring'), the adapter is only found if it is a
                        suffix of the read.
  -g ADAPTER, --front=ADAPTER
                        Sequence of an adapter ligated to the 5' end (paired
                        data: of the first read). The adapter and any
                        preceding bases are trimmed. Partial matches at the
5'
                        end are allowed. If a '^' character is prepended
                        ('anchoring'), the adapter is only found if it is a
                        prefix of the read.


  Additional read modifications:
    -u LENGTH, --cut=LENGTH
                          Remove bases from each read (first read only if
                          paired). If LENGTH is positive, remove bases from
the
                          beginning. If LENGTH is negative, remove bases from
                          the end. Can be used twice if LENGTHs have
different
                          signs.
    --nextseq-trim=3'CUTOFF
                          NextSeq-specific quality trimming (each read).
Trims
                          also dark cycles appearing as high-quality G bases
                          (EXPERIMENTAL).
    -q [5'CUTOFF,]3'CUTOFF, --quality-cutoff=[5'CUTOFF,]3'CUTOFF
                          Trim low-quality bases from 5' and/or 3' ends of
each
                          read before adapter removal. Applied to both reads
if
                          data is paired. If one value is given, only the 3'
end
                          is trimmed. If two comma-separated cutoffs are
given,
                          the 5' end is trimmed with the first cutoff, the 3'
                          end with the second.
    --quality-base=QUALITY_BASE
                          Assume that quality values in FASTQ are encoded as
                          ascii(quality + QUALITY_BASE). This needs to be set
to
                          64 for some old Illumina FASTQ files. Default: 33
    -l LENGTH, --length=LENGTH
                          Shorten reads to LENGTH. This and the following
                          modificationsare applied after adapter trimming.
    --trim-n              Trim N's on ends of reads.
    --length-tag=TAG      Search for TAG followed by a decimal number in the
                          description field of the read. Replace the decimal
                          number with the correct length of the trimmed read.
                          For example, use --length-tag 'length=' to correct
                          fields like 'length=123'.
```

Below is an example of how you can use cutadapt on your data. Note that you cannot call cutadapt without telling the shell where executable file is. To do so, load conda environment with `conda activate env_NGS`

```
# Change directory
cd "${st_path}"/course_materials/fastq/

# Activate Conda environment
conda deactivate
conda activate env_NGS

# Cutadapt manual
cutadapt --help

#######################################################
#  Sample multiplexing using DNA barcodes
#######################################################

# Cutadapt run and interpret the output
# NOTE: I have split the command into lines for better
# visibility. You need the back slash immediately followed by end of line
# to indicate to the shell that the command continues on the next line.
# If you put the command in one line, remove the back slashes.

cutadapt -g Positive=^GATACA \
        -g Negative=^AGTAGT \
        -g BQ=^CACACA \
        -g Long=^AAACCC \
        -o trimmed_'{name}'.fq final_merge_synthetic_reads.fq
```

Your cutadapt output should look like more or less like this:

```
This is cutadapt 1.18 with Python 2.7.15
Command line parameters: -g Positive=^GATACA -g Negative=^AGTAGT -g BQ=^CACACA -g Long=^AAACCC -o trimmed-{name}.fq merge_synteti
c_reads.fq
Processing reads on 1 core in single-end mode ...
Finished in 59.35 s (17 us/read; 3.63 M reads/minute).

=== Summary ===

Total reads processed:               3,587,730
Reads with adapters:                 3,587,730 (100.0%)
Reads written (passing filters):     3,587,730 (100.0%)

Total basepairs processed:   256,163,640 bp
Total written (filtered):    234,637,260 bp (91.6%)

=== Adapter Positive ===

Sequence: GATACA; Type: anchored 5'; Length: 6; Trimmed: 1076320 times.

No. of allowed errors:
0-6 bp: 0

Overview of removed sequences
length  count   expect  max.err error counts
6       1076320 875.9   0       1076320

=== Adapter Negative ===

Sequence: AGTAGT; Type: anchored 5'; Length: 6; Trimmed: 1076320 times.

No. of allowed errors:
0-6 bp: 0

Overview of removed sequences
length  count   expect  max.err error counts
6       1076320 875.9   0       1076320

=== Adapter BQ ===

Sequence: CACACA; Type: anchored 5'; Length: 6; Trimmed: 1076320 times.

No. of allowed errors:
0-6 bp: 0

Overview of removed sequences
length  count   expect  max.err error counts
6       1076320 875.9   0       1076320

=== Adapter Long ===

Sequence: AAACCC; Type: anchored 5'; Length: 6; Trimmed: 358770 times.

No. of allowed errors:
0-6 bp: 0

Overview of removed sequences
length  count   expect  max.err error counts
6       358770  875.9   0       358770
```

5. Summarising statistics with MultiQC

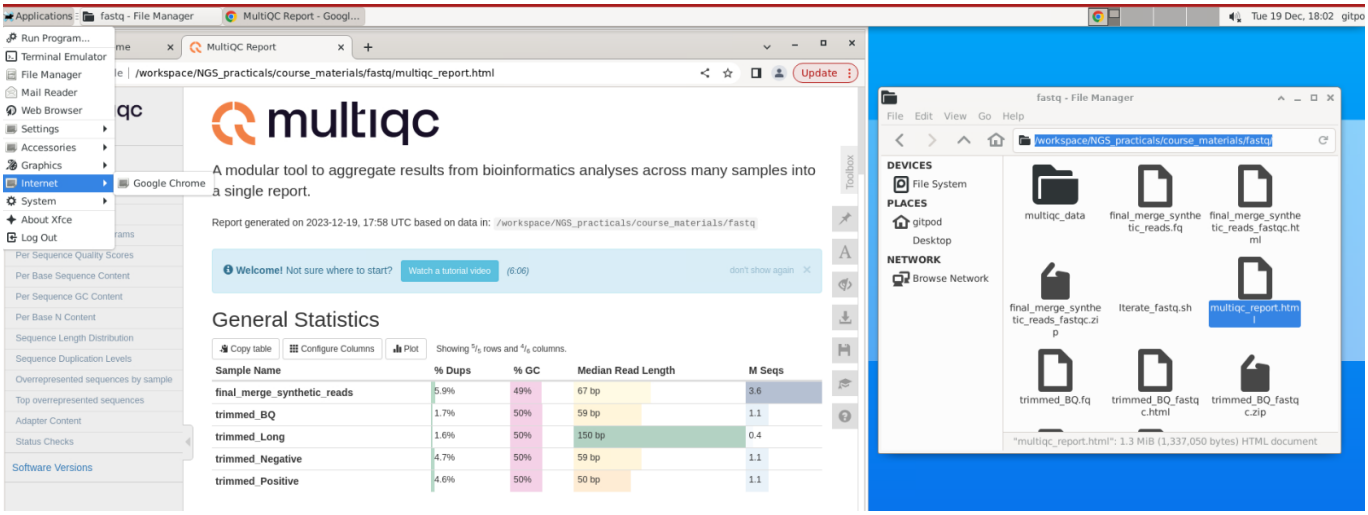
MultiQC is a very useful program that allows you to summarise statistics that have been produced by a number of other programs. A modular tool to aggregate results from bioinformatics analyses across many samples into a single report. In this part of the practical, we are summarising the **FastQC** output for the merged fastq file, as well as the statistics output by **Cutadapt** after demultiplex or trimming of the reads was carried out.

https://multiqc.info/

```
# Conda deactivate previous Conda environments.
conda deactivate

# Create and load conda environment with all the tools needed get MultiQC
results.
# mamba env create -f "${st_path}"/env/.environment_multiqc.yaml
conda activate env_multiqc
```

```
# manual
multiqc --help

#####################################################
# Summarising statistics with MultiQC
#####################################################
# run multiQC.
# "." dot means in the current directory and downstream.
# "-f" removing previous results.
multiqc . -f

# Open in Firefox
firefox multiqc_report.html
```

If this doesn't work you could navigate with the folder system File System > Workspace > NGS_practicals > course_materials > fastq ... and open "multiqc_report.html" with Firefox or any other web browser. To select a Web browser, open applications tab on the top left corner and select Internet > Google Chrome. Or just drag and drop the multiqc-report to Chrome.



6. Automating repetitive tasks with bash scripts

Simple bash script

We can create simple scripts on bash to help us and perform routinely tasks. As a follow on, we also runs a **bash shell script** that carries out **FastQC** quality control on each of the demultiplexed samples - separated out from the merged raw data - and re-runs **MultiQ**C at the end on all these outputs. Inspect the "Iterate_fastq.sh" script and learn how you should loop over files in bash.

```bash
#!/bin/bash

# Assign everyting ended in .fq to FILES variable
FILES=*.fq

# Loop over FILES and assing each to file
for file in $FILES
do
    filename=$(basename "$file")
    extension="${filename##*.}"
    filename="${filename%.*}"

    echo "File on the loop:             $file"
    echo "Curated base name of file:    $filename"
    echo "Extension:                    $extension"

    # Call FastQC quality analysis
    /s/software/fastqc/v0.11.8/FastQC/fastqc ${file}

    echo -e "######################\n\n"
done

# Run MultiQC
# -f    Overwrite any existing reports
#  .    on this directory
echo "Running MultiQC..."
echo "Capturing fastQC statistics"
/s/software/anaconda/python3/bin/multiqc -f .
```

Shebang First line to tell which interpreter to use

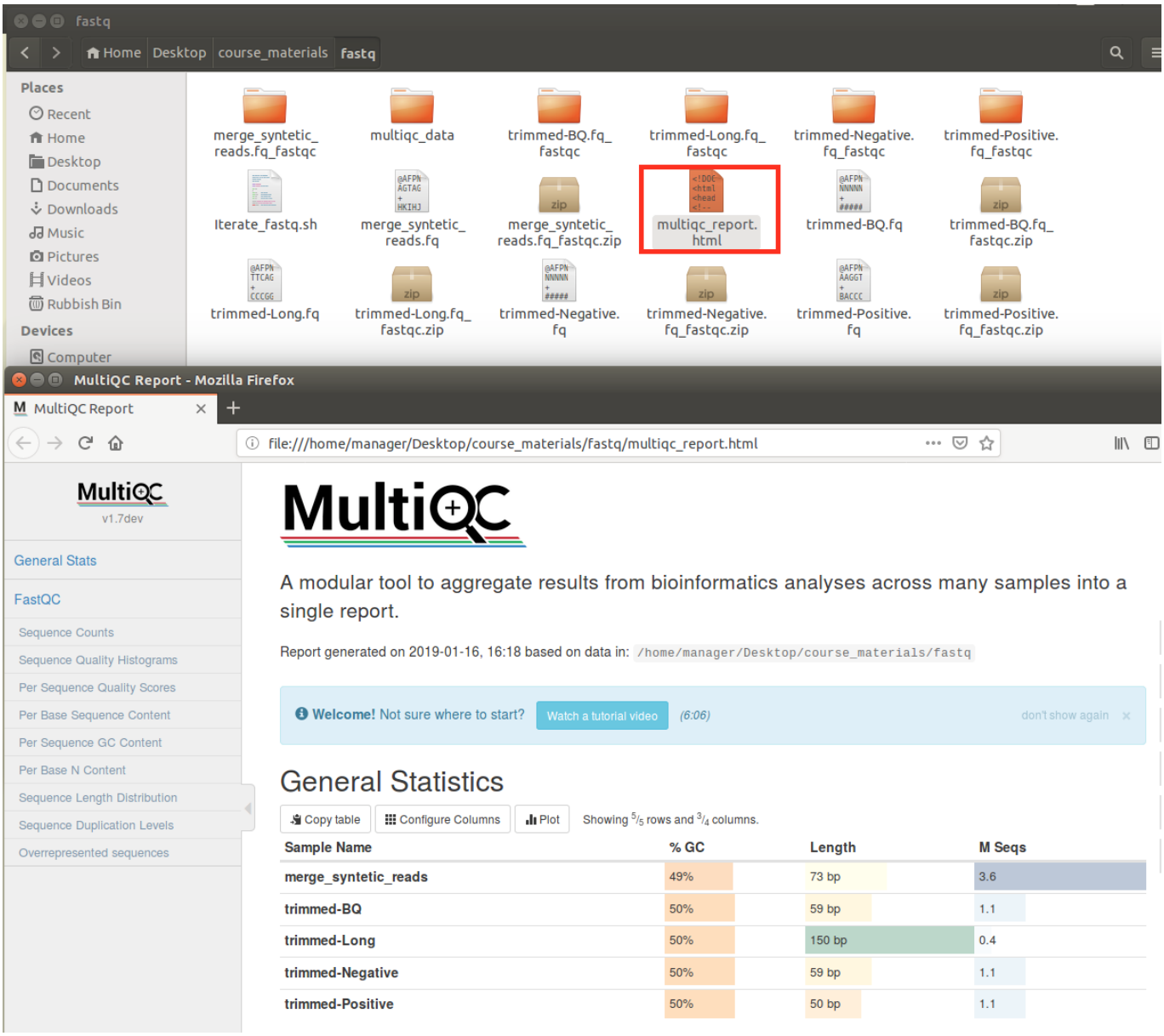
for i in $list
do
done

bash for loop

Running simple bash script

```
# Create and Load conda environment with all the tools needed to carry on
this practical.
conda deactivate
conda activate env_multiqc

# Go to folder that stores all Fastq files
cd "${st_path}"/course_materials/fastq/

# Copy resource script
cp /workspace/NGS_practicals/course_materials/resources/Iterate_fastq.sh .

# Inspect simple shell script
less Iterate_fastq.sh

# To iterate FastQC program over all the .fastq files on the current folder
and run multiQC wrapper that will search and parse all the NGS output
files.
bash Iterate_fastq.sh

# Open multiple FastQC reports on Firefox. Check multiQC report !!! NOTE
Close firefox window or "ctr + c" on the command line.
firefox multiqc_report.html

# Deactivate environment
conda deactivate
```

Your multiQC output should look something like this (top of the html page):



7. Aligning reads to a reference genome with Bowtie2

We have quality-inspected, demultiplexed, quality-inspected again our reads and now finally we arrive to the point when we can align those reads to a reference genome, that of E. coli.

A commonly used program for aligning short reads to a reference genome is **bowtie2**. There are many more such programs: bwa, STAR, hisat2 etc. The new generation of sequencing that produces much longer reads uses its own set of aligners that are more suitable for much longer sequences and sequences that generally also contain a higher error rate compared with Illumina's short reads.

In this practical, we use Illumina short reads and we will map them using bowtie2. This program is installed by anaconda, but the latest version can be run by loading Conda `env_NGS` environment.

Bowtie2 manual

Information about bowtie2 can be found here: http://bowtie-bio.sourceforge.net/bowtie2/manual.shtml

You can also check the help function of bowtie2 available as is customary with the --help argument:

```
# Activate environment
conda deactivate
conda activate env_NGS

# Bowtie2 manual
bowtie2 --help
```

```
Bowtie2
Usage:
  bowtie2 [options]* -x <bt2-idx> {-1 <m1> -2 <m2> | -U <r>} [-S <sam>]

  <bt2-idx>  Index filename prefix (minus trailing .X.bt2).
             NOTE: Bowtie 1 and Bowtie 2 indexes are not compatible.
  <m1>       Files with #1 mates, paired with files in <m2>.
             Could be gzip'ed (extension: .gz) or bzip2'ed (extension:
.bz2).
  <m2>       Files with #2 mates, paired with files in <m1>.
             Could be gzip'ed (extension: .gz) or bzip2'ed (extension:
.bz2).
  <r>        Files with unpaired reads.
             Could be gzip'ed (extension: .gz) or bzip2'ed (extension:
.bz2).
  <sam>      File for SAM output (default: stdout)

  <m1>, <m2>, <r> can be comma-separated lists (no whitespace) and can be
  specified many times.  E.g. '-U file1.fq,file2.fq -U file3.fq'.

Options (defaults in parentheses):
 Input:
  -s/--skip <int>    skip the first <int> reads/pairs in the input (none)
  -u/--upto <int>    stop after first <int> reads/pairs (no limit)
  -5/--trim5 <int>   trim <int> bases from 5'/left end of reads (0)
  -3/--trim3 <int>   trim <int> bases from 3'/right end of reads (0)

Alignment:
  -N <int>           max # mismatches in seed alignment; can be 0 or 1 (0)
  -L <int>           length of seed substrings; must be >3, <32 (22)
  -i <func>          interval between seed substrings w/r/t read len
(S,1,1.15)
  ...
  --end-to-end       entire read must align; no clipping (on)
   OR
  --local            local alignment; ends might be soft clipped (off)

 Presets:                 Same as:
  For --end-to-end:
   --very-fast            -D 5 -R 1 -N 0 -L 22 -i S,0,2.50
   --fast                 -D 10 -R 2 -N 0 -L 22 -i S,0,2.50
   --sensitive            -D 15 -R 2 -N 0 -L 22 -i S,1,1.15 (default)
   --very-sensitive       -D 20 -R 3 -N 0 -L 20 -i S,1,0.50
```

```
  For --local:
   --very-fast-local      -D 5 -R 1 -N 0 -L 25 -i S,1,2.00
   --fast-local           -D 10 -R 2 -N 0 -L 22 -i S,1,1.75
   --sensitive-local      -D 15 -R 2 -N 0 -L 20 -i S,1,0.75 (default)
   --very-sensitive-local -D 20 -R 3 -N 0 -L 20 -i S,1,0.50

 Reporting:
  (default)          look for multiple alignments, report best, with MAPQ
   OR
  -k <int>           report up to <int> alns per read; MAPQ not meaningful
   OR
  -a/--all           report all alignments; very slow, MAPQ not meaningful
```

Understanding local vs end-to-end alignment

Reference: http://gensoft.pasteur.fr/docs/bowtie2/2.1.0/#end-to-end-alignment-versus-local-alignment

Similarly to global vs local alignments you came across in the Sequence Analysis and Genomics module, bowtie2 can carry out alignments that are either "end-to-end" or "local". See the examples below for the meaning of these two types of alignment in the context of bowtie2.

```
~End-to-end alignment example~

The following is an "end-to-end" alignment because it involves all the
characters in the read. Such an alignment can be produced by Bowtie2 in
either end-to-end mode or in local mode.

Read:      GACTGGGCGATCTCGACTTCG
Reference: GACTGCGATCTCGACATCG

Alignment:
  Read:      GACTGGGCGATCTCGACTTCG
             |||||  |||||||||| |||
  Reference: GACTG--CGATCTCGACATCG

~Local alignment example~

The following is a "local" alignment because some of the characters at the
ends of the read do not participate. In this case, 4 characters are omitted
(or "soft trimmed" or "soft clipped") from the beginning and 3 characters
are omitted from the end. This sort of alignment can be produced by Bowtie
2 only in local mode.

Read:      ACGGTTGCGTTAATCCGCCACG
Reference: TAACTTGCGTTAAATCCGCCTGG

Alignment:
  Read:      ACGGTTGCGTTAA-TCCGCCACG
```

```
                  |||||||| ||||||
  Reference: TAACTTGCGTTAAATCCGCCTGG
```

Create Bowtie2 genome index

In general, aligners like bowtie2 require the indexing of the reference genome in order to speed up the alignment process (remember: there are millions of short sequences to align against potentially billions of letters in a genome). Bowtie2 is no exception. Thus, the first step before aligning the reads is to index the reference genome. The code below shows how this is done.

```
# Change directory
cd "${st_path}"/course_materials/genomes/AFPN02.1

# Create Bowtie2 genome index
bowtie2-build AFPN02.1_merge.fasta AFPN02.1_merge
```

If you built the index successfully, you should end up with several .bt2 files in the directory course_materials/genomes/AFPN02.1/.

Aligning sequencing reads to the reference genome using Bowtie2.

The steps below align the separated and trimmed reads to the reference genome, one file at a time.

Create a directory to keep the results of the alignment This directory will be created under your $st_path directory

```
# Change and create directory
cd "${st_path}"/course_materials/
mkdir results_NGS1
cd "${st_path}"/course_materials/results_NGS1

#####################################################
# Aligning with Bowtie2 the four demultiplexed files
#####################################################

# Align with Bowtie2
# Align trimmed_Long.fq reads
# --all           # Report all alignments; very slow, MAPQ not meaningful
# --end-to-end    # Align with end-to-end setting
# -x              # Bowtie2 index path
# -U              # fastq file path
# -S              # Output as sam file
# 2>              # Redirect error output to file in bash

# Align trimmed_BQ.fq reads
```

```
bowtie2 --all --end-to-end -x
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge -U
"${st_path}"/course_materials/fastq/trimmed_BQ.fq -S BQ.sam 2>
BQ_bowtie_stats.txt

# Align trimmed_Long.fq reads
bowtie2 --all --end-to-end -x
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge -U
"${st_path}"/course_materials/fastq/trimmed_Long.fq -S Long.sam 2>
Long_bowtie_stats.txt

# Align trimmed_Negative.fq
bowtie2 --all --end-to-end -x
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge -U
"${st_path}"/course_materials/fastq/trimmed_Negative.fq -S Negative.sam 2>
Negative_bowtie_stats.txt

# Align trimmed_Positive.fq reads
bowtie2 --all --end-to-end -x
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge -U
"${st_path}"/course_materials/fastq/trimmed_Positive.fq -S Positive.sam 2>
Positive_bowtie_stats.txt
```

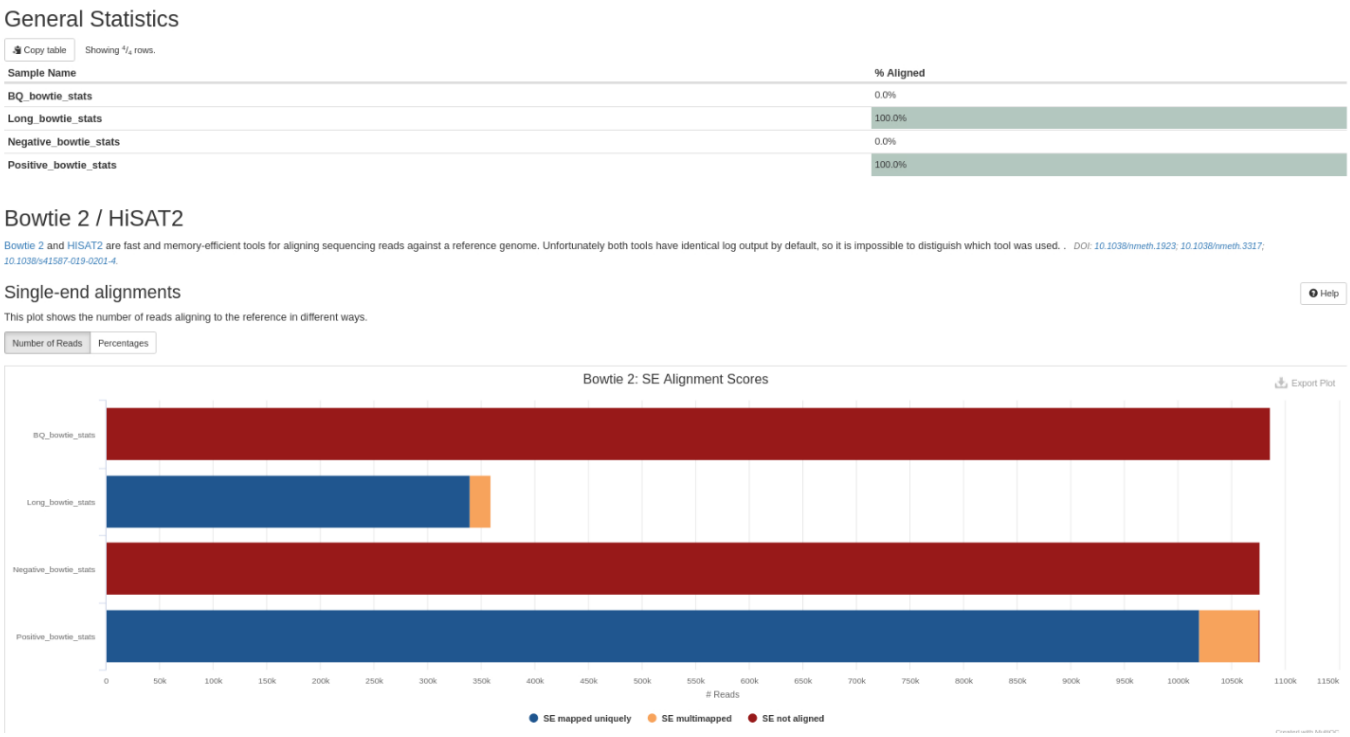
Explore Bowtie2 alignment results.

Do you see differences between the statistic of aligned positive and negative samples?

```
ls -l
head Positive_bowtie_stats.txt
head Negative_bowtie_stats.txt
```

run MultiQC to capture all the Bowtie2 alignment statistics.

```
conda activate env_multiqc
multiqc . -f

# Open in Firefox and explore mapping results.
# Are you able to map negative or BQ samples?
# Why? Check fastQC statistics of negative or BQ samples in comparison with
Positive or Long samples.
# Can you improve mapping trimming reads using previous steps? or even
using Bowtie2?
firefox multiqc_report.html
```



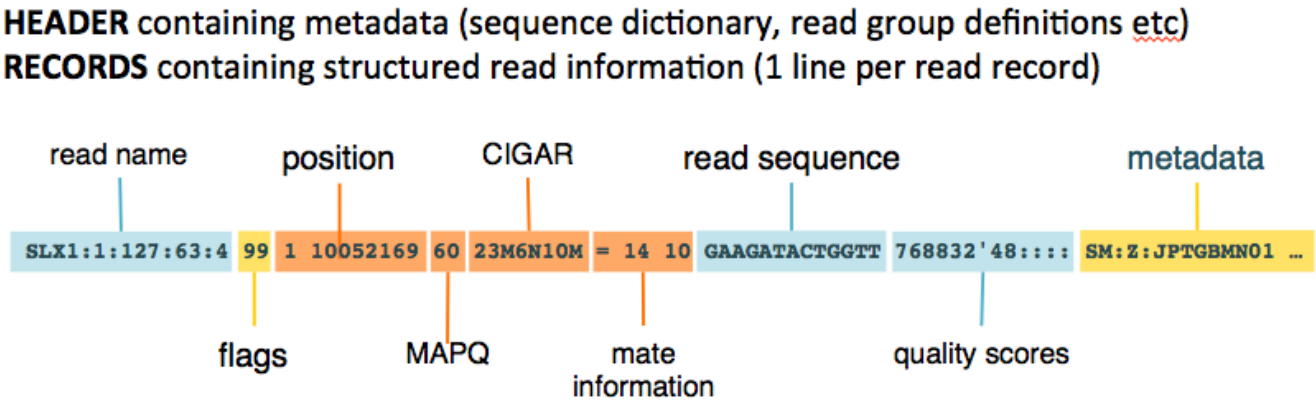
Think about the following questions:

Are you able to map negative or BQ samples? If not, why not? Check fastQC statistics of negative or BQ samples and compare them with Positive or Long samples. Can you improve the mapping by trimming reads or by using Bowtie2 alone?

8. SAM Sequence Alignment/Map format

Programs that map NGS reads produce SAM formatted files; this is a format where each read is shown together with information on where it was mapped, if it was mapped at all and with what quality scores each base in the sequence was mapped. Often, these SAM-formatted files are zipped in a special way that produces a binary format equivalent to SAM, known as BAM. You can view and edit SAM files with normal text editors and shell commands that can be used on text files but you cannot view BAM files this way. This is where programs like samtools are handy. samtools is a program that allows you to view and manipulate easily SAM and BAM files.

You can read more about the SAM specification here: https://samtools.github.io/hts-specs/SAMv1.pdf . You can see that your files have several lines starting with @ which represent the header section,followed by one line per read, which represent the records in the SAM file. The image below shows the parts of a SAM record:



Lets explore the aligned ".sam" files using the simple head or less command:

Explanation of sam header. **Try to identify read name, flags, position, MAPQ, CIGAR, read sequence, quality scores and metadata in your data**. Can you see any difference in the reads coming from Positive and negative samples? What they have at the 5' and 3' ends?

Do you see differences between aligned Positive and Negative samples?

```
#######################################################
# Explore sam flat text aligned reads output
#######################################################

head -n 10 Positive.sam
head -n 10 Negative.sam
```

```
thoth.cryst.bbk.ac.uk> head -n 10 Positive.sam
@HD     VN:1.0  SO:unsorted
@SQ     SN:AFPN02.1_merge       LN:5381621
@PG     ID:bowtie2      PN:bowtie2      VN:2.3.4.3      CL:"/s/software/anaconda/python3/bin/bowtie2-align-s --wrapper basic-0 --end-to-end -x /d/projects/u/ri001/course_materials/genomes/AFPN02.1/AFPN02.1_
merge -q /d/projects/u/ri001/course_materials/fastq/trimmed_Positive.fq -S Positive.sam"
AFPN02.1_merge-1076320  0       AFPN02.1_merge  2202341 42      50M     *       0       0       AAGGTCGCTAATCTCTTTACGCAGATTTTTTATTCCTTCAACTAACAGCG      BACCCGFGGGGGGGGGG1GGEGGG0GBGG1GGGGGGGG<GGGGGGGGBCG   A
S:i:0   XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i:0  MD:Z:50 YT:Z:UU
AFPN02.1_merge-1076319  0       AFPN02.1_merge  1795375 42      50M     *       0       0       CTTTATCCCGAAAGCGTTTGGTAGCTCGCTGGCATTAACGGGTTCGCCAG      BBCCB=G>GGGGGE>GGGGGGG@GGGGDGGGGGGGFGGGGGGGGG@GGGG   A
S:i:0   XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i:0  MD:Z:50 YT:Z:UU
AFPN02.1_merge-1076318  16      AFPN02.1_merge  4286405 42      50M     *       0       0       GGTGCTGGAGATGATCACCCGCCTCGCGCCGAACGCGGAAATCGGCAAGA      G0G/GGDGBG@GGCGGE1GGG>0GGCGGGGGGFGGGGGGGGGGGDCCCCB   A
S:i:0   XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i:0  MD:Z:50 YT:Z:UU
AFPN02.1_merge-1076317  0       AFPN02.1_merge  4824345 42      50M     *       0       0       TCACCGAGAAGAAGATATGTAATTCGCCAGAGAGCGACGGCCCTTGCTGA      BCBCCGGG/GG1GGGGGGGFGGGGGGG<GGDEG1GGGGGGEG>GGGGGGG   A
S:i:-3  XN:i:0  XM:i:1  XO:i:0  XG:i:0  NM:i:1  MD:Z:8C41       YT:Z:UU
AFPN02.1_merge-1076316  16      AFPN02.1_merge  1706195 42      50M     *       0       0       TCATGATCACCACAATCGATACCAGCGACAACAGCAGCGAGATAGTAAAC      G1GGGGGGGDGGGGGGFGGGG@GGGGGGGGGGGD;FCGFGGGGG;C=C3C   A
S:i:0   XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i:0  MD:Z:50 YT:Z:UU
AFPN02.1_merge-1076315  16      AFPN02.1_merge  964540  42      50M     *       0       0       ATACCGTTACGCCAGGCGAGCAGCCAGACCAACAGAGAAGTGACAATGCC      GGGGEGGGGGGGGGG1GCFGGGGGGGGGGGGGG>EGGGG1GGGGGG@CCCC   A
S:i:0   XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i:0  MD:Z:50 YT:Z:UU
AFPN02.1_merge-1076314  16      AFPN02.1_merge  3690848 42      50M     *       0       0       TTACCTCAATGCTCTGAATTTTAGCGCTGGTTTCAAGAAGGATACTTTCT      GGGGGGGGGG<GGGGGGGGGGGGGGGGGG/GGGGGGGGGGGGGFG0CBCC   A
S:i:0   XN:i:0  XM:i:0  XO:i:0  XG:i:0  NM:i:0  MD:Z:50 YT:Z:UU
thoth.cryst.bbk.ac.uk> head -n 10 Negative.sam
@HD     VN:1.0  SO:unsorted
@SQ     SN:AFPN02.1_merge       LN:5381621
@PG     ID:bowtie2      PN:bowtie2      VN:2.3.4.3      CL:"/s/software/anaconda/python3/bin/bowtie2-align-s --wrapper basic-0 --end-to-end -x /d/projects/u/ri001/course_materials/genomes/AFPN02.1/AFPN02.1_
merge -q /d/projects/u/ri001/course_materials/fastq/trimmed_Negative.fq -S Negative.sam"
AFPN02.1_merge-1076320  4       *       0       0       *       *       0       0       NNNNNAAGGTCGCTAATCTCTTTACGCAGATTTTTTATTCCTTCAACTAACAGCGNNNN     #####BACCCGFGGGGGGGGGG1GGEGGG0GBGG1GGGGGGGG<GGGGGGGGBC
G####   YT:Z:UU YF:Z:NS
AFPN02.1_merge-1076319  4       *       0       0       *       *       0       0       NNNNNCTTTATCCCGAAAGCGTTTGGTAGCTCGCTGGCATTAACGGGTTCGCCAGNNNN     #####BBCCB=G>GGGGGE>GGGGGGG@GGGGDGGGGGGGFGGGGGGGGG@GGG
G####   YT:Z:UU YF:Z:NS
AFPN02.1_merge-1076318  4       *       0       0       *       *       0       0       NNNNNTCTTGCCGATTTCCGCGTTCGGCGCGAGGCGGGTGATCATCTCCAGCACCNNNN     #####BCCCCDGGGGGGGGGGGFGGGGGGCGG0>GGG1EGGCGG@GBGDGG/G0
G####   YT:Z:UU YF:Z:NS
AFPN02.1_merge-1076317  4       *       0       0       *       *       0       0       NNNNNTCACCGAGAAGAAGATATGTAATTCGCCAGAGAGCGACGGCCCTTGCTGANNNN     #####BCBCCGGG/GG1GGGGGGGFGGGGGGG<GGDEG1GGGGGGEG>GGGGGG
G####   YT:Z:UU YF:Z:NS
AFPN02.1_merge-1076316  4       *       0       0       *       *       0       0       NNNNNGTTTACTATCTCGCTGCTGTTGTCGCTGGTATCGATTGTGGTGATCATGANNNN     #####C3C=C;GGGGGFGCF;DGGGGGGGGGGG@GGGGFGGGGGGDGGGGGGG1
G####   YT:Z:UU YF:Z:NS
AFPN02.1_merge-1076315  4       *       0       0       *       *       0       0       NNNNNGGCATTGTCACTTCTCTGTTGGTCTGGCTGCTCGCCTGGCGTAACGGTATNNNN     #####CCCC@GGGGGG1GGGGE>GGGGGGGGGGGGGGGFCG1GGGGGGGGGEGGG
G####   YT:Z:UU YF:Z:NS
AFPN02.1_merge-1076314  4       *       0       0       *       *       0       0       NNNNNAGAAAGTATCCTTCTTGAAACCAGCGCTAAAATTCAGAGCATTGAGGTAANNNN     #####CCBC0GFGGGGGGGGGGGGG/GGGGGGGGGGGGGGGGGG<GGGGGGGGG
G####   YT:Z:UU YF:Z:NS
```

Exploring samtools further

The manual for **samtools**, a suite of programs for exploring high-throughput sequencing data in SAM/BAM formats is here:

http://www.htslib.org/doc/samtools.html

Try out the code below to explore samtools - each of the commands following samtools is a keyword directing samtools towards a different functionality.

```
# Samtools manual
conda deactivate
conda activate env_NGS
samtools --help


# samtools FILTER reads manual
samtools view --help

# samtools sort manual
samtools sort --help

# samtools stats manual
samtools stats --help
```

"Samtools view" main usage:

```
samtools view main usage
Usage: samtools view [options] <in.bam>|<in.sam>|<in.cram> [region ...]

Options:
  -b       output BAM
  -C       output CRAM (requires -T)
  -1       use fast BAM compression (implies -b)
  -u       uncompressed BAM output (implies -b)
  -h       include header in SAM output
  -H       print SAM header only (no alignments)
  -c       print only the count of matching records
  -o FILE  output file name [stdout]
  -U FILE  output reads not selected by filters to FILE [null]
  -t FILE  FILE listing reference names and lengths (see long help) [null]
  -L FILE  only include reads overlapping this BED FILE [null]
  -r STR   only include reads in read group STR [null]
  -R FILE  only include reads with read group listed in FILE [null]
  -q INT   only include reads with mapping quality >= INT [0]
  -l STR   only include reads in library STR [null]
  -m INT   only include reads with number of CIGAR operations consuming
           query sequence >= INT [0]
  -f INT   only include reads with all  of the FLAGs in INT present [0]
  -F INT   only include reads with none of the FLAGS in INT present [0]
  -G INT   only EXCLUDE reads with all  of the FLAGs in INT present [0]
  -s FLOAT subsample reads (given INT.FRAC option value, 0.FRAC is the
           fraction of templates/read pairs to keep; INT part sets seed)
```

```
  -x STR   read tag to strip (repeatable) [null]
  -B       collapse the backward CIGAR operation
  -?       print long help, including note about region specification
  -S       ignored (input format is auto-detected)
      --input-fmt-option OPT[=VAL]
               Specify a single input file format option in the form
               of OPTION or OPTION=VALUE
  -O, --output-fmt FORMAT[,OPT[=VAL]]...
               Specify output format (SAM, BAM, CRAM)
      --output-fmt-option OPT[=VAL]
               Specify a single output file format option in the form
               of OPTION or OPTION=VALUE
  -T, --reference FILE
               Reference sequence FASTA FILE [null]
  -@, --threads INT
               Number of additional threads to use [0]
```

Manipulate aligned data in the sam format

The code below transforms sam to bam (a compressed format of sam), indexes the bam file (required by many programs, e.g. genome viewers) and retrieves some more statistics of aligned sequences using samtools. NOTE: you could rewrite this as a script that runs through all the .sam files and carries out the same process on each one. **Samtools**

```
cd "${st_path}"/course_materials/results_NGS1/
ls -l

#####################################################
# Samtools aligned data exploration
#####################################################
# BQ.sam
# sort and compress
samtools sort BQ.sam > BQ.bam
# create index
samtools index BQ.bam
# alignment statistics
samtools stats BQ.bam > BQ_stats.txt
# simple statistics
samtools flagstat BQ.bam > BQ_flagstat.txt


# Long.sam
samtools sort Long.sam > Long.bam
samtools index Long.bam
samtools stats Long.bam > Long_stats.txt
samtools flagstat Long.bam > Long_flagstat.txt


# Negative.sam
```

```
samtools sort Negative.sam > Negative.bam
samtools index Negative.bam
samtools stats Negative.bam > Negative_stats.txt
samtools flagstat Negative.bam > Negative_flagstat.txt


# Positive.sam
samtools sort Positive.sam > Positive.bam
samtools index Positive.bam
samtools stats Positive.bam > Positive_stats.txt
samtools flagstat Positive.bam > Positive_flagstat.txt
```

Capture all the statistic and data produced during this practical using MultiQC

```
# Activate environment
conda deactivate
conda activate env_multiqc

# Change directory
cd "${st_path}"/course_materials/

#####################################################
# Final MultiQC capturing all the data
#####################################################
# Run MultiQC on the initial folder
multiqc . -f

# Inspect MultiQC report
firefox multiqc_report.html
```

9. View aligned reads on a genome viewer

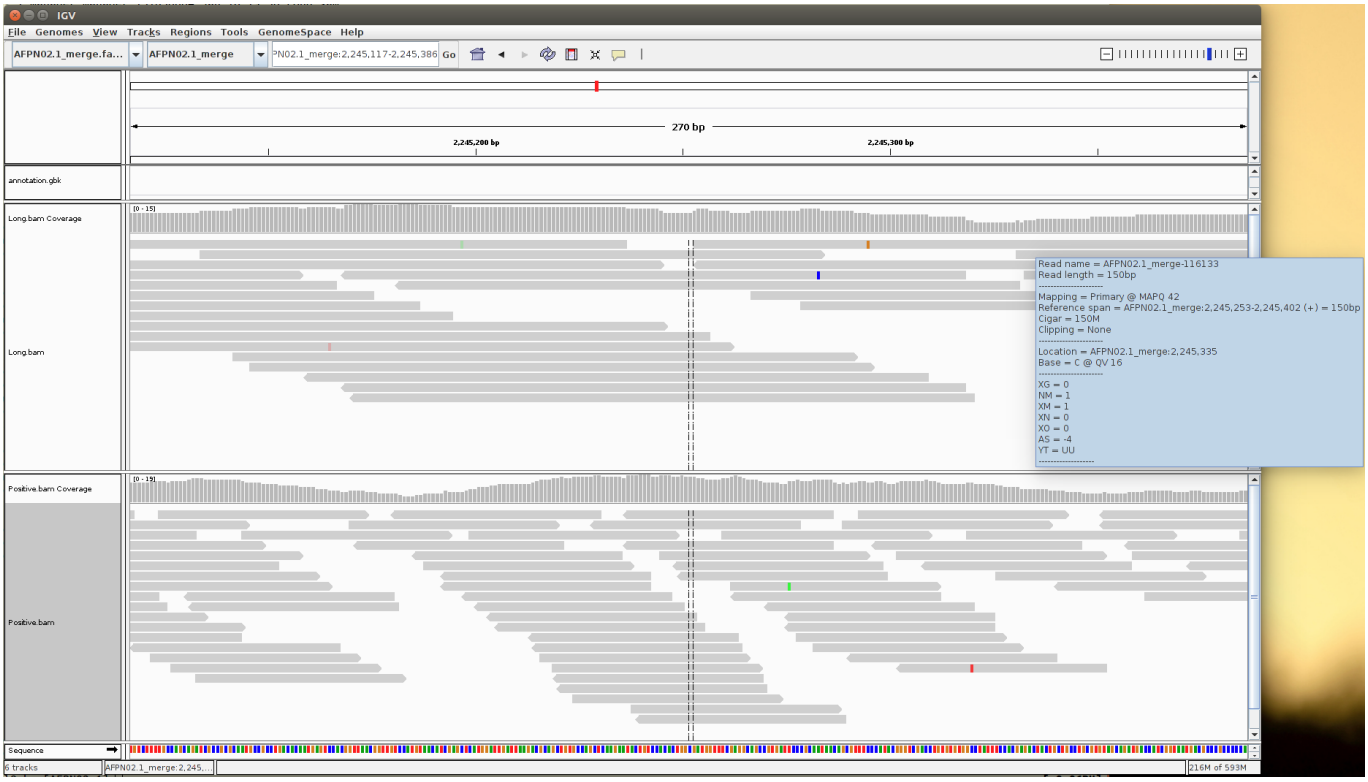
The Integrative Genomics Viewer (IGV) is a high-performance visualization tool for interactive exploration of large, integrated genomic datasets. It supports a wide variety of data types, including array-based and next-generation sequence data, and genomic annotations.

Open IGV program by typing `igv`. Load genome Genomes > load genome from file > select path `"${st_path}"/course_materials/genomes/AFPN02.1" and fasta sequence "AFPN02.1.fasta"`.

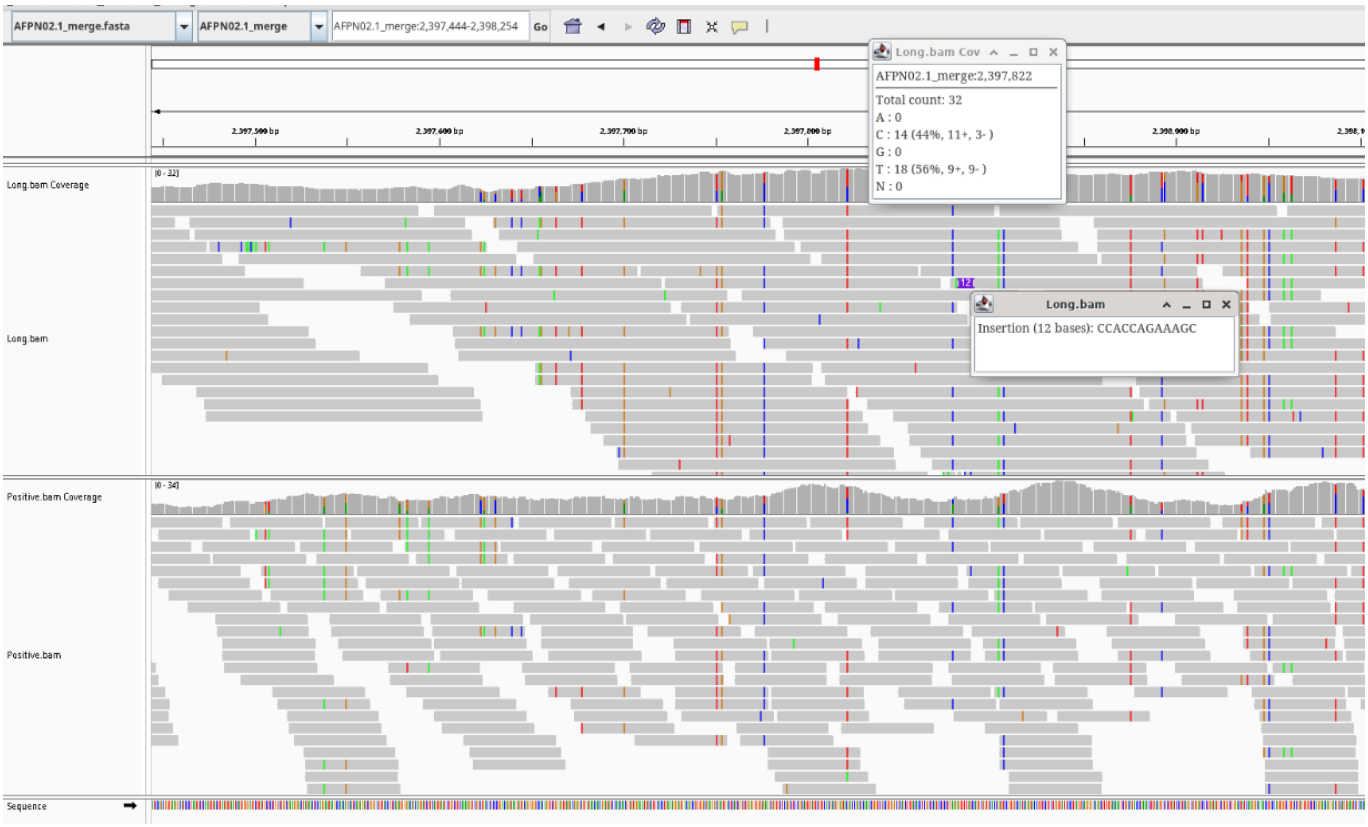
Open bam files file > load from file > select path `"${st_path}"/course_materials/results_NGS1/` and all the `*.bam files`. NOTE that you will need the bam file to be indexed with samtools (`*.bam.bai`). sam_format

```
# Activate environment
conda deactivate
conda activate env_NGS
```

```
# Load genome reference file "AFPN02.1_merge.fasta"
# Go to genome locus "AFPN02.1_merge:2397587"
# Load Long and Positive aligned data to IGV.
igv --genome
/workspace/NGS_practicals/course_materials/genomes/AFPN02.1/AFPN02.1_merge.
fasta /workspace/NGS_practicals/course_materials/results_NGS1/Long.bam --
locus "AFPN02.1_merge:2397587"
/workspace/NGS_practicals/course_materials/results_NGS1/Positive.bam
```



Browse aligned reads region `AFPN02.1_merge:22397537-2399083`



10. Get consensus sequence with freebayes

Very easy to run for simple SNP calling. Does not assume any ploidy.

```
cd "${st_path}"/course_materials/results_NGS1/

#####################################################
# Get consensus sequence merging all the variants
#####################################################
freebayes -f
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge.fasta -p 1
Long.bam > Long.vcffile
vcf2fasta -f
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge.fasta -P 1
Long.vcffile -p AFPN02.1_consensus.fasta

# Rename file to more precise name
mv AFPN02.1_consensus.fastaunknown_AFPN02.1_merge:0.fa
AFPN02.1_merge_consensus.fa

# Read VCF file
less Long.vcffile

# Read fasta file
less AFPN02.1_merge_consensus.fa

# alternative not used
# samtools mpileup -uf
"${st_path}"/course_materials/genomes/AFPN02.1/AFPN02.1_merge.fasta
Long.bam | bcftools view -cg | vcfutils.pl vcf2fq > CONSENSUS.fq
```